

***WOMEN AS POLICY MAKERS: A PHILOSOPHY OF
SCIENCE ANALYSIS OF A RANDOMIZED POLICY
EXPERIMENT***

Popper, Quine–Duhem, and Kuhn applied to
Chattopadhyay & Duflo (2004)

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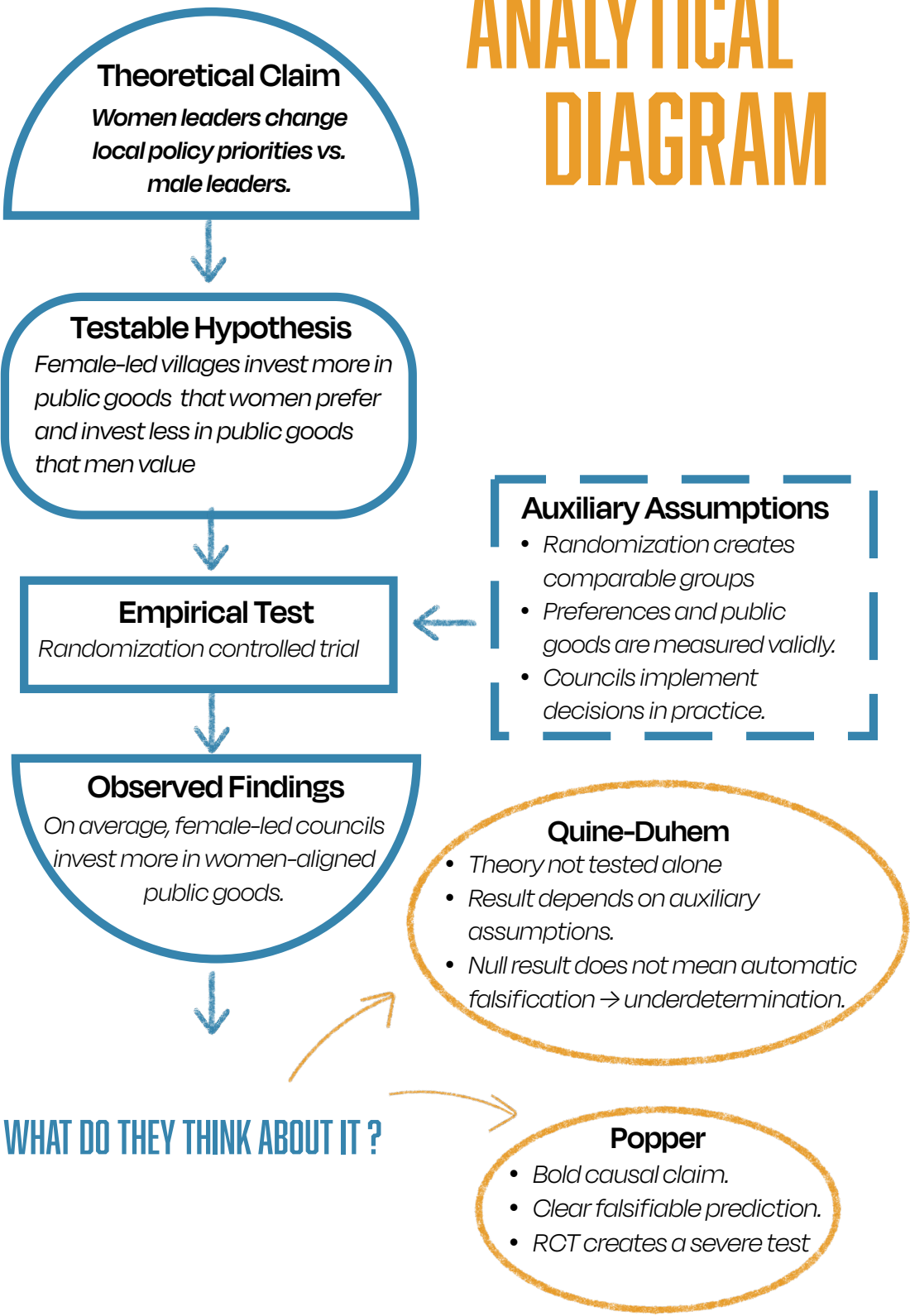
PHILOSOPHY OF SCIENCE ANALYSIS : WOMEN AS POLICY MAKERS IN INDIA

RESEARCH QUESTION : DOES LEADER GENDER CHANGE POLICY PRIORITIES ?

TEST METHOD

- Random assignment of village councils to female leaders via seat reservations.
- Use female-led and male-led villages for comparison
- Compare public goods investments across female-led vs. male-led councils.

ANALYTICAL DIAGRAM



RESULTS COMPARISON

	Theoretical prediction	Findings
Investment in women-preferred goods	<u>Increases under female leadership</u>	<u>Yes it increases</u>
Investment in men-preferred goods	<u>Does not increase under female leadership</u>	<u>Does not increase</u>

PHILOSOPHICAL INTERPRETATIONS

POPPER : IS THIS GOOD SCIENCE ?

FALSIFIABILITY PRINCIPLE

Good science must be testable + provable wrong by observations

FALSIFYING OBSERVATION

- No difference between female/male villages
- OR
- Investment pattern is unrelated to women preferences

It would prove the causal claim wrong

RCT = STRONG TEST

- Random assignment SO fair comparison
- It rules out alternatives explanations

QUINE-DUHEM: TESTED ALONE?

CORE IDEA

Hypotheses never tested alone, always with auxiliary assumptions

2 KEY ASSUMPTIONS

1. Reservation truly random → comparable villages pre-treatment
2. Complaints validly measure preferences

WHAT IF NO EFFECTS ARE FOUND ??

Blame theory OR blame the assumptions?

Impossible to know = **UNDERDETERMINATION**
Same evidence fits multiple theories . Data cannot choose what to reject

KUHN: NORMAL SCIENCE OR ANOMALY?

WHAT IS A PARADIGM ?

Shared rules + methods that scientists accept without questioning. (Ex: RCT = best way to find causes)

METHOD = Normal Science

Study uses standard RCT tools:

- Random assignment of female leaders
- Balance checks + robustness tests

THEORY = CREATES PROBLEMS

Dominant models say: leader gender shouldn't change policy (Downs/Coase)

BUT study shows: female leaders DO change policy → ANOMALY!

VERDICT : NORMAL METHOD BUT PROBLEMATIC THEORY

SUMMARY

Popper: Falsifiable causal claim; randomization strengthens the test.

Quine–Duhem: Evidence depends on auxiliary assumptions (randomization, measurement, implementation).

Kuhn: Normal-science method; findings challenge models where leader identity should not matter.

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